HARDWARE SYSTEMS ENGINEERING | PCB DESIGN & EMBEDDED INTEGRATION

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OVERVIEW

Electronics student (7th semester, UAA) building toward complete hardware systems, not isolated pieces: PCB design, embedded firmware, and the digital logic underneath. Designed and routed boards from schematic to fabrication-ready Gerbers, deployed a production hardware/firmware/software safety-interlock system at a food manufacturing plant, and validated a power supply design across SPICE, KiCad, and COMSOL thermal FEM. Computer architecture background (RISC-V CPU model, synthesized on real FPGAs) that still informs how I approach hardware, no longer the main focus.

TECHNICAL SKILLS

PCB & Power Electronics
KiCad (schematic to fab-ready Gerbers), LTspice, COMSOL FEM (thermal), SCR / buck / Class-D power stages, linear regulator design

Embedded Systems
STM32, FreeRTOS, RFID, ARM Cortex-M4, custom drivers, cross-compiling (ARM GCC toolchain)

HDL & FPGA
Verilog (RTL), VHDL, Xilinx ISE, Gowin EDA, Amaranth HDL

Programming & Debug
C, Python, ARM Assembly, C++, Git, OpenOCD/GDB, GTKWave, Linux (Arch/Manjaro, 6+ yrs daily use)

EDUCATION

Bachelor of Science in Electronics Engineering
Universidad Autónoma de Aguascalientes (UAA) • 7th Semester • GPA 9.5/10 • Cambridge English B2 Certified

KEY PROJECTS

IoT Safety Checklist & Hardware Interlock System
Raspberry Pi / Node.js / Shelly Smart Relay
IKGB105/IOT-Safety-Checklist
Production system deployed at a food manufacturing plant. Digitizes pre-shift machine safety checklists and physically interlocks machine startup — a smart relay wired into the start-button circuit only closes after the checklist is approved. Runs fully offline as its own WiFi access point, web server, and REST API, with photo evidence capture, automated WhatsApp alerts, and an audited supervisor-override flow. Strongest evidence of full-stack integration: hardware, firmware, and software, actually running, not a class exercise.

ARM Processor PCB Design
KiCad, 2-Layer PCB, ARM Cortex-M (QFP)
IKGB105/PCB-ARM-Kicad
Designed and routed a 2-layer PCB from schematic capture through fabrication-ready Gerbers and drill files, around an ARM Cortex-M MCU with USB power/programming, user I/O, and expansion headers for BOOT mode selection and a camera interface. Used a ground pour across the stackup to keep return paths short without a dedicated internal ground plane.

Adjustable LM317 Power Supply — SPICE, PCB & Thermal FEM
LTspice, KiCad, COMSOL Multiphysics
IKGB105/LM317-Adjustable-PowerSupply
End-to-end power supply design documented as an engineering logbook: LTspice simulation of the adjustable topology, KiCad PCB layout, and a COMSOL thermal FEM study of the regulator's worst-case junction temperature that drove two rounds of heatsink selection. Simulation caught a real 76 mV arithmetic error in an earlier hand-calculation before it propagated further. Companion LM2596 buck converter root-caused its thermal limit to the catch diode, not the regulator.

IK-Dron — 5" FPV Racing Drone
BetaFlight, SpeedyBee F405 V4
IKGB105/IK-Dron
Team project: diagnosed and flew a 5" racing drone after its flight controller suffered an unrecoverable hardware failure. Migrated to a SpeedyBee F405 V4 stack on BetaFlight, then root-caused a takeoff failure via the CLI to a propeller/motor-rotation mismatch tripping BetaFlight's runaway-takeoff protection.

LANGUAGES

Spanish — Native

English — B2 (Cambridge Certified)